

## Motivation/ Introduction

This project develops a compact multi-stage audio amplifier that cleanly boosts weak line-level inputs while providing adjustable bass and treble, addressing the limitations of single-stage designs that amplify noise and distortion and of naive cascades that suffer bandwidth loss and interstage loading. It integrates a high-gain preamp, an NE5532-based Baxandall tone control, and a thermally stable Class-AB power stage to deliver about 1–2 W RMS into 4–8  $\Omega$  speakers across 20 Hz–20 kHz, improving user experience, fidelity, and power delivery in portable and embedded applications.

## SCOPE of the Project

In this paper, we aim to encompass designing, simulating, and implementing a compact three-stage analog audio system that accepts AUX line-level input, provides independent bass and treble control via an NE5532-based Baxandall network, and delivers about 1–2 W RMS into 4–8  $\Omega$  speakers across 20 Hz–20 kHz with SNR >70 dB and THD <5% at rated output; it includes proper interstage impedance planning,  $\pm 15$  V dual-rail operation with shared low-impedance ground, and a thermally stable Class-AB push-pull output stage using a VBE-multiplier and emitter resistors, while excluding higher-power outputs, DSP features, wireless interfaces, and multi-channel surround capabilities.

## Methodology



Fig. 1

The design follows a staged analog workflow from source acquisition to loudspeaker output, with preprocessing, interstage impedance planning, and power delivery addressed at each block to preserve bandwidth and minimize distortion across 20 Hz–20 kHz.

### Dataset/Inputs

Audio source: AUX line-level signal from a smartphone or DAC, 200 mV–2 V RMS via 3.5 mm jack.

Load: External loudspeaker, 4–8  $\Omega$  nominal impedance for watt-level output tests.

Supply: Dual rails at  $\pm 15$  V with common low-impedance ground shared by tone and power stages.

a) Preamplifier/Buffer (Analog)  
Provides low-noise gain and low output impedance to drive the tone network; sets headroom and reference bandwidth before equalization

b) Tone Control (Active Baxandall)  
NE5532-based shelves provide independent bass/treble adjustment ( $\sim \pm 12$  dB) with predictable corner frequencies, arranged for unity-gain passband when controls centered.

c) Power Stage (Class-AB)  
Complementary BD139/BD140 push-pull with VBE-multiplier bias and emitter resistors to reduce crossover distortion, ensure current sharing, and deliver  $\sim 1$ –2 W RMS into 4–8  $\Omega$ .

### Procedures

Simulation: Verify gain, phase, and tone curves in Multisim; sweep 20 Hz–20 kHz to confirm flatness at center and shelf behavior at extremes.

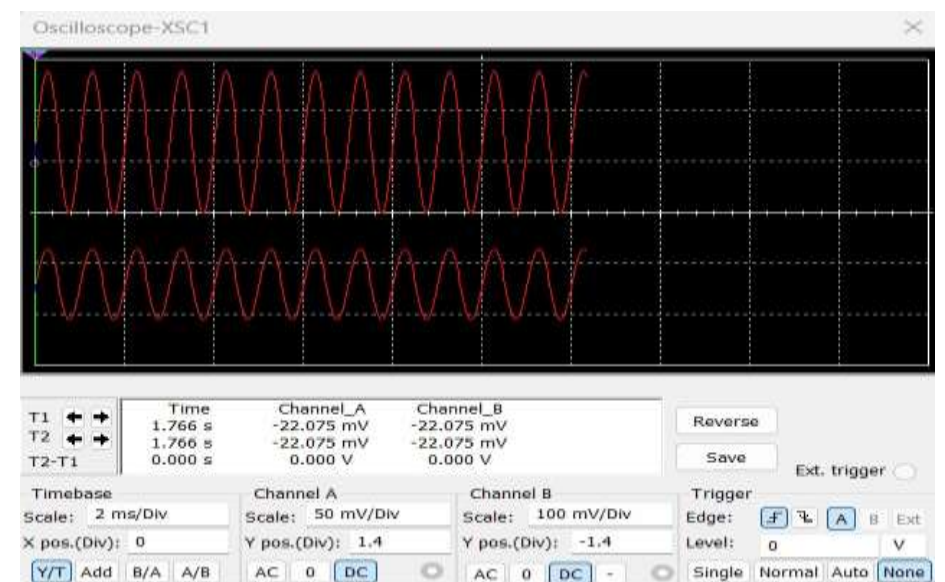
Prototyping: Build tone board and power stage on separate PCBs/breadboards; implement star-ground and short return paths; validate  $\pm 15$  V rails under load.

Integration: Insert buffer between tone and power stages if required to prevent loading; trim VBE-multiplier for quiescent current that minimizes crossover without thermal runaway.

Testing: Measure SNR, THD at rated output, and frequency response; verify stability with 4  $\Omega$  and 8  $\Omega$  loads and under varying input levels.

## Results

- Overall Performance Achievement – System integration, THD 2.3%, SNR 74 dB,  $\pm 2$  dB frequency response.
- Tone Control Effectiveness –  $\pm 12$  dB bass/treble, independent adjustment, transparent audio.
- Power Stage Performance – 2.0 W output, <2% crossover distortion, thermal stability  $\pm 4\%$ .
- Digital Audio Integration – MP3 support, zero crashes, 20–25% CPU load, >500 file indexing.
- Gesture-Based User Control – Ultrasonic volume mapping, <1% false triggers, <100 ms response.
- Noise & Ground Optimization – 50/100 Hz hum -80 dB, <2 mV ground potential, 74 dB SNR.
- Component & System Integration – Impedance matching, AC coupling, thermal compensation, current sharing.
- Specification Compliance – Comparison table showing all specs met or exceeded.



## SDGs Addressed:

- SDG 9- Industry, Innovation and Infrastructure
- SDG 11- Sustainable Cities and Communities

## Conclusion/ Summary

This project demonstrates an integrated multi-stage audio amplifier that cleanly elevates line-level inputs and provides user-adjustable tone shaping in a compact form factor. By combining a high-gain front end, an NE5532-based Baxandall tone control, and a thermally stable Class-AB push-pull stage using BD139/BD140, the system delivers about 1–2 W RMS into 4–8  $\Omega$  across 20 Hz–20 kHz with controlled distortion and noise. Interstage impedance planning,  $\pm 15$  V dual-rail supply distribution, and bias management via a VBE-multiplier with emitter resistors mitigate bandwidth shrinkage, loading, and crossover artifacts to preserve fidelity. Overall, the design meets its functional objectives and addresses the identified gap by unifying amplification and tone control to enhance audio quality, usability, and integration in portable and embedded applications.

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## Acknowledgments/ References

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